KAIZEN MATHS

Lesson starter

Linear Algebra: Systems, Span, Basis and Transformations

Linear Transformations, Kernel and Image

Name _____

S1

Find the standard matrix and use it to transform the vector.

A linear transformation has $T\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}-2\\1\end{bmatrix}$ and $T\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}4\\4\end{bmatrix}$. Find the standard matrix and $T\begin{bmatrix}-2\\4\end{bmatrix}$.

S2

Find the kernel of each singular matrix.

Find the kernel of $A = \begin{bmatrix}-2 & -4\\4 & 8\end{bmatrix}$.

S3

Find a basis for the image and state the rank.

Find a basis for the image of $A = \begin{bmatrix}1 & 0 & 1\\-3 & 4 & 1\end{bmatrix}$ and state the rank.

S4

Find the rank, nullity, and kernel.

For the row-reduced matrix $R = \begin{bmatrix}1 & 0 & -1\\0 & 1 & 3\end{bmatrix}$, find the rank, nullity, and kernel of the associated map from R^3 to R^2.

Show your working

KAIZEN MATHS

Lesson starter

Linear Algebra: Systems, Span, Basis and Transformations

Linear Transformations, Kernel and Image

Name _____

S1

Find the standard matrix and use it to transform the vector.

A linear transformation has $T\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}-2\\1\end{bmatrix}$ and $T\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}4\\4\end{bmatrix}$. Find the standard matrix and $T\begin{bmatrix}-2\\4\end{bmatrix}$.

S2

Find the kernel of each singular matrix.

Find the kernel of $A = \begin{bmatrix}-2 & -4\\4 & 8\end{bmatrix}$.

S3

Find a basis for the image and state the rank.

Find a basis for the image of $A = \begin{bmatrix}1 & 0 & 1\\-3 & 4 & 1\end{bmatrix}$ and state the rank.

S4

Find the rank, nullity, and kernel.

For the row-reduced matrix $R = \begin{bmatrix}1 & 0 & -1\\0 & 1 & 3\end{bmatrix}$, find the rank, nullity, and kernel of the associated map from R^3 to R^2.

ALG-E2AE-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Linear Transformations, Kernel and Image

Name _____

EA1

Find the standard matrix and use it to transform the vector.

A linear transformation has $T\begin{pmatrix}\begin{bmatrix}1\\0\end{bmatrix}\end{pmatrix} = \begin{bmatrix}3\\4\end{bmatrix}$ and $T\begin{pmatrix}\begin{bmatrix}0\\1\end{bmatrix}\end{pmatrix} = \begin{bmatrix}3\\-4\end{bmatrix}$. Find the standard matrix and $T\begin{pmatrix}\begin{bmatrix}3\\3\end{bmatrix}\end{pmatrix}$.

EA2

Find the kernel of each singular matrix.

Find the kernel of $A = \begin{bmatrix}-3 & -2\\9 & 6\end{bmatrix}$.

Show your working

KAIZEN MATHS

Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Linear Transformations, Kernel and Image

Name _____

EB1

Find a basis for the image and state the rank.

Find a basis for the image of $A = \begin{bmatrix}1 & 0 & 1\\2 & 1 & 3\end{bmatrix}$ and state the rank.

Show your working

KAIZEN MATHS

Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Linear Transformations, Kernel and Image

Name _____

EA1

Find the standard matrix and use it to transform the vector.

A linear transformation has $T\begin{pmatrix}\begin{bmatrix}1\\0\end{bmatrix}\end{pmatrix} = \begin{bmatrix}3\\4\end{bmatrix}$ and $T\begin{pmatrix}\begin{bmatrix}0\\1\end{bmatrix}\end{pmatrix} = \begin{bmatrix}3\\-4\end{bmatrix}$. Find the standard matrix and $T\begin{pmatrix}\begin{bmatrix}3\\3\end{bmatrix}\end{pmatrix}$.

EA2

Find the kernel of each singular matrix.

Find the kernel of $A = \begin{bmatrix}-3 & -2\\9 & 6\end{bmatrix}$.

ALG-E2AE-L3 · EA

Show your working

KAIZEN MATHS

Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Linear Transformations, Kernel and Image

Name _____

EB1

Find a basis for the image and state the rank.

Find a basis for the image of $A = \begin{bmatrix}1 & 0 & 1\\2 & 1 & 3\end{bmatrix}$ and state the rank.

ALG-E2AE-L3 · EB

Show your working

Teacher answers & working

Linear Algebra: Systems, Span, Basis and Transformations · Linear Transformations, Kernel and Image

ALG-E2AE-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Find the standard matrix and use it to transform the vector.

A linear transformation has $T\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}-2\\1\end{bmatrix}$ and $T\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}4\\4\end{bmatrix}$. Find the standard matrix and $T\begin{bmatrix}-2\\4\end{bmatrix}$.

Answer:

$A = \begin{bmatrix}-2 & 4\\1 & 4\end{bmatrix}$, $T\begin{bmatrix}-2\\4\end{bmatrix} = \begin{bmatrix}20\\14\end{bmatrix}$

Working:

The columns of the standard matrix are the images of the standard basis vectors.

$A = \begin{bmatrix}-2 & 4\\1 & 4\end{bmatrix}$.

Now multiply A by the vector.

$$\begin{bmatrix}-2 & 4\\1 & 4\end{bmatrix}\begin{bmatrix}-2\\4\end{bmatrix} = \begin{bmatrix}-2(-2)+16\\1(-2)+16\end{bmatrix} = \begin{bmatrix}20\\14\end{bmatrix}.$$

S2

Find the kernel of each singular matrix.

Find the kernel of $A = \begin{bmatrix}-2 & -4\\4 & 8\end{bmatrix}$.

Answer:

$\ker(A) = \text{span}\{\begin{bmatrix}-4\\2\end{bmatrix}\}$

Working:

Solve $A\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$.

$-2x - 4y = 0$ and $4x + 8y = 0$.

The second equation is −2 times the first, so only $-2x - 4y = 0$ is needed.

Choose $x = -4t$ and $y = 2t$. Then $-2(-4t) - 8t = 0$.

$\begin{bmatrix}x\\y\end{bmatrix} = t\begin{bmatrix}-4\\2\end{bmatrix}$, so $\ker(A) = \text{span}\{\begin{bmatrix}-4\\2\end{bmatrix}\}$.

Teacher answers & working

Linear Algebra: Systems, Span, Basis and Transformations · Linear Transformations, Kernel and Image

ALG-E2AE-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Find a basis for the image and state the rank.

Find a basis for the image of A = $\begin{bmatrix} 1 & 0 & 1 \\ -3 & 4 & 1 \end{bmatrix}$ and state the rank.

Answer:

Image basis: $\{\begin{bmatrix} 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \end{bmatrix}\}$. Rank = 2.

Working:

The image is spanned by the columns of A.

Column 1 is $\begin{bmatrix} 1 \\ -3 \end{bmatrix}$, column 2 is $\begin{bmatrix} 0 \\ 4 \end{bmatrix}$, and column 3 is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Column 3 = column 1 + column 2, so it does not add a new direction.

Columns 1 and 2 are not scalar multiples of each other, so they are independent.

A basis for the image is $\{\begin{bmatrix} 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \end{bmatrix}\}$ and the rank is 2.

S4

Find the rank, nullity, and kernel.

For the row-reduced matrix R = $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 3 \end{bmatrix}$, find the rank, nullity, and kernel of the associated map from R^3 to R^2.

Answer:

rank = 2, nullity = 1, ker = span$\{\begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix}\}$

Working:

There are pivots in columns 1 and 2, so rank = 2.

There are 3 columns in total, so nullity = 3 − 2 = 1.

Solve R $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

$x - z = 0$, so $x = z$.

$y + 3z = 0$, so $y = -3z$.

Let z = t. Then $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t\begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix}$.

Teacher answers & working

Linear Algebra: Systems, Span, Basis and Transformations · Linear Transformations, Kernel and Image

ALG-E2AE-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EA1

Find the standard matrix and use it to transform the vector.

A linear transformation has T($\begin{bmatrix}1\\0\end{bmatrix}$) = $\begin{bmatrix}3\\4\end{bmatrix}$ and T($\begin{bmatrix}0\\1\end{bmatrix}$) = $\begin{bmatrix}3\\-4\end{bmatrix}$. Find the standard matrix and T($\begin{bmatrix}3\\3\end{bmatrix}$).

Answer:

$A = \begin{bmatrix}3 & 3\\4 & -4\end{bmatrix}$, T($\begin{bmatrix}3\\3\end{bmatrix}$) = $\begin{bmatrix}18\\0\end{bmatrix}$

Working:

The columns of the standard matrix are the images of the standard basis vectors.

$A = \begin{bmatrix}3 & 3\\4 & -4\end{bmatrix}$.

Now multiply A by the vector.

$\begin{bmatrix}3 & 3\\4 & -4\end{bmatrix}\begin{bmatrix}3\\3\end{bmatrix} = \begin{bmatrix}3(3)+9\\4(3)-12\end{bmatrix} = \begin{bmatrix}18\\0\end{bmatrix}$.

EA2

Find the kernel of each singular matrix.

Find the kernel of A = $\begin{bmatrix}-3 & -2\\9 & 6\end{bmatrix}$.

Answer:

ker(A) = span{$\begin{bmatrix}-2\\3\end{bmatrix}$}

Working:

Solve A$\begin{bmatrix}x\\y\end{bmatrix}$ = $\begin{bmatrix}0\\0\end{bmatrix}$.

$-3x - 2y = 0$ and $9x + 6y = 0$.

The second equation is −3 times the first, so only −3x − 2y = 0 is needed.

Choose x = −2t and y = 3t. Then −3(−2t) − 6t = 0.

$\begin{bmatrix}x\\y\end{bmatrix}$ = t$\begin{bmatrix}-2\\3\end{bmatrix}$, so ker(A) = span{$\begin{bmatrix}-2\\3\end{bmatrix}$}.

Teacher answers & working

Linear Algebra: Systems, Span, Basis and Transformations · Linear Transformations, Kernel and Image

ALG-E2AE-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EB1

Find a basis for the image and state the rank.

Find a basis for the image of $A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix}$ and state the rank.

Answer:

Image basis: $\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\}$. Rank = 2.

Working:

The image is spanned by the columns of A.

Column 1 is $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, column 2 is $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and column 3 is $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$.

Column 3 = column 1 + column 2, so it does not add a new direction.

Columns 1 and 2 are not scalar multiples of each other, so they are independent.

A basis for the image is $\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\}$ and the rank is 2.